

A COMPARISON OF NORWAY RATS AND SPINY MICE ON MEASURES OF ACTIVITY AND EXPLORATION

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Summary.—In Exp. 1 6 adult male Long-Evans derived rats (*Rattus norvegicus*) and 6 adult male spiny mice (*Acomys cahirinus*) were observed individually inside a 60-cm × 30-cm × 30-cm open field during a 15-min. session. Measures of locomotion, immobility, self-grooming, rearing-up, and object contact were obtained. Tests of species differences indicated that spiny mice showed significantly more locomotor and rearing behavior than Norway rats. In Exp. 2 12 adult Norway rats (6 male, 6 female) and 12 adult spiny mice (6 male, 6 female) were observed individually inside a six-compartment maze during a 15-min. session. Although no sex difference in maze exploration was found, spiny mice showed significantly more maze exploration than Norway rats. These results suggest that spiny mice, highly precocial in development relative to Norway rats, exhibit correlates of that neonatal competence as adults evidenced by greater behavioral activity and locomotor exploration in novel environments.

Norway rats (*Rattus norvegicus*) and spiny mice (*Acomys cahirinus*) represent different extremes on the continuum of rate of pre-adult development. Spiny mice give birth to precocial offspring (Porter & Doane, 1978). The neonates are covered with short coarse hairs, sensory modalities appear functional, and they are able to locomote independently at birth. From laboratory breeding data, median litter size has been reported at three offspring (Porter & Wyrick, 1979). Norway rats by comparison demonstrate altricial development characterized by less motor and sensory capacity at birth, large litter size, and slower rate of locomotor development during pre-weaning.

The descriptive and comparative studies reported here sought to identify, in a preliminary way, relationships between adult behavior and the rate-of-development dimension represented by the differences between these two species during the pre-weaning period. Exp. 1 provides measurements of open-field activity and movement by adults of each species. Exp. 2 yields measurements of movement through a multi-compartment exploration maze by adults of each species. The general objective of these studies was to determine if representative precocial and altricial rodents, demonstrating activity and locomotor differences as neonates, would show similar differences as adults. If so, the generality of the precocial-altricial continuum across age levels would be confirmed for this comparison.

GENERAL METHOD

Long-Evans derived rats (*Rattus norvegicus*) were obtained from a

¹We thank Richard H. Porter for help in establishing our breeding colony of spiny mice. Address reprint requests to Richard Deni, Psychology Department, Rider College, Lawrenceville, New Jersey 08648.

laboratory breeding colony. Rats were housed in litters with the dam until weaning, followed by a period of group-housing with littermates until approximately 50 days of age. Rats were next housed individually. Testing began at approximately 60 days of age.

Spiny mice (*Acomys cahirinus*) were obtained from a laboratory breeding colony. Mice were housed with their dam and sire until weaning. Mice were then housed in male-female pairs. Testing began at approximately 60 days of age.

Both Norway rats and spiny mice were maintained with Purina 5012 chow and water freely available.

EXP. 1: COMPARISON OF MOVEMENT AND ACTIVITY

Six adult male rats and six adult male spiny mice were tested in a 60-cm $\times$ 30-cm $\times$ 30-cm Plexiglas open field containing an empty glass water bottle located on the floor in the center of the enclosure. The floor was marked with a grid of 18 10-cm square movement zones.

Subjects were observed individually, during a single 15-min. session. Measures of movement included the number of grid zones traversed, the duration of immobility, and the frequency of rearing-up on hind legs. Measures of individual activity included duration of self-grooming and number of times the subject contacted the glass bottle in the center of the field.

Table 1 shows the means and standard deviations for each measure in both species. Spiny mice showed more locomotion, less immobility, more rearing, less grooming, and more object contact than Norway rats. Analyses of variance to test species differences showed that spiny mice locomoted and reared significantly more than rats.

These data document species differences between spiny mice and Norway rats in the baseline occurrence of locomotor and individual behaviors in a novel environment. Furthermore, these differences are correlated with the

TABLE 1
MEANS AND STANDARD DEVIATIONS FOR MEASURES OF OPEN-FIELD BEHAVIOR BY
NORWAY RATS AND SPINY MICE (NUMBER OF OPEN-FIELD ZONES ENTERED)

Group		<i>M</i>	<i>SD</i>	<i>F</i> †
Locomotion	Norway rats	313.8	121.1	19.6†
	Spiny mice	665.8	152.5	
Immobility	Norway rats	59.4	26.8	0.85
	Spiny mice	70.9	48.6	
Rearing	Norway rats	98.3	175.8	11.77*
	Spiny mice	38.2	40.0	
Self-grooming	Norway rats	138.9	41.9	3.98
	Spiny mice	117.3	20.3	
Contacts	Norway rats	45.7	63.2	1.7
	Spiny mice	20.5	25.6	

†One-way between-subjects analysis of variance: Locomotion, rearing, contacts with objects (*f*); Immobility, self-grooming (*sec.*). **df* = 1/10, *p* < .01. †*df* = 1/10, *p* < .005.

rate-of-development dimension (precocial-altricial) for motor and sensory capacity in these two species, i.e., newborn spiny mice are precocial in sensorimotor functioning.

Exp. 2 was designed to demonstrate that the species differences in locomotor behavior in Exp. 1 reflected differences in exploratory tendency.

EXP. 2: EXPLORATION OF A MULTI-COMPARTMENT MAZE

Twelve adult Norway rats (6 male, 6 female) and 12 adult spiny mice (6 male, 6 female) ran in a multi-compartment maze to assess exploratory behavior. The maze consisted of a 90-cm $\times$ 45-cm $\times$ 15-cm Plexiglas enclosure divided into six compartments by a series of offset partitions mounted perpendicular to the long sides of the enclosure. To traverse the maze, subjects had to zig-zag around these partitions. Each of the six 15-cm $\times$ 45-cm compartments was marked with a grid of three 15-cm $\times$ 15-cm movement zones used to score locomotion inside a given compartment.

Each subject was observed individually inside the maze during a single 15-min. session. Exploration was scored based on a count of movement zones entered per compartment per session. Subjects were introduced into the maze through an aperture on one of the 45-cm sides. Movement through the six compartments thus represented progressive exploration away from the start.

Table 2 shows the means and standard deviations for exploration scores (number of zones entered) as a function of species and maze compartment. Species, sex, and compartment differences were tested using a mixed three-factor analysis of variance with species and sex as between-subjects factors and compartments as a within-subjects factor. Results indicated that spiny mice showed significantly more exploration than rats ($F_{1,20} = 35.9$, $p < .001$). No significant sex difference was found. There was a significant difference in exploration as a function of maze compartment (see Table 2; $F_{5,100} = 20.3$, $p < .001$). *Post hoc* Newman-Keul's test results for the maze compartment effect showed that the second and third compartments were associated with significantly higher exploration scores (regardless of species) compared

TABLE 2
MEANS AND STANDARD DEVIATIONS FOR NUMBER OF COMPARTMENTS ENTERED BY
NORWAY RATS AND SPINY MICE AS A FUNCTION OF SPECIES
AND COMPARTMENT IN A COMPLEX MAZE

Group		<i>M</i>	<i>SD</i>
Species	Norway rats	40.2	15.6
	Spiny mice	78.4	24.0
Maze	One (Close to Start)	60.2	25.9
	Two	72.8	30.0
	Three	63.4	29.5
	Four	57.4	26.8
	Five	60.3	26.7
	Six	41.8	20.6

with Compartment six, most remote from the start box ($p < .05$). Exploration in Compartment six was one-third lower on average than that exhibited in any of the first three compartments. One of the four possible interaction effects, the interaction of species by sex by compartment, was significant ($F_{5,100} = 3.6, p < .005$). *Post hoc* tests showed that this interaction reflected consistent differences as a function of species (spiny mice scored higher) at all combinations of sex and compartment.

GENERAL DISCUSSION

Both Exps. 1 and 2 suggest a connection between the adult behavior and activity level of spiny mice and the high rate of physical and behavioral development in these animals during the pre-weaning period. The precocial spiny mice were more active and exploratory in laboratory tests than altricial Norway rats. However, the fact that these two species differ in so many ways requires a limited interpretation of the present preliminary findings.

Previous work with spiny mice has supported the hypothesis that neonatal precocity correlates with adult behavior. Precocity in the young coincides with evidence of extended maternal responsiveness and care (Porter & Doane, 1978), a connection which strengthens the relationship hypothesized between small litter size, active and developmentally accelerated offspring, and parental investment.

Activity and exploratory behavior in rodents correlate with a variety of factors. For example, inbred Norway rat strains show less exploratory tendency than wild-caught animals (Price & Huck, 1976). Additional ecological factors affecting activity would include among others, habitat, resource utilization, and predation (cf. Kaufman, 1974). However, an explanation of these activity and exploration data in spiny mice using habitat factors is hampered by the lack of formal observations made in the species-typical Near East arid environment. Available information (R. H. Porter, personal communication) suggests that spiny mouse habitations are shallow, dispersed, and employ available cover (quite different from the extensive communal burrow system of Norway rats). Species differences of this type raise additional questions concerning the underlying bases for differences in activity and exploration.

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